AI + HEALTH ASSISTANT

Because better health begins with a conversation...

Your Health Companion on a Call

A Voice-Enabled Health Risk Assessment System using Machine Learning, Retrieval-Augmented Generation and Vapi

FastAPI | scikit-learn / XGBoost | LangChain | FAISS | Groq | Vapi

How It Works ?

1 You Make a Call
Call the AI Health Assistant and start a natural conversation.



2 Answer a Few Questions
The assistant asks key questions about your lifestyle and medical history.



3 Get Your Health Insights
Your inputs are analysed using a Machine Learning model and your question is answered using a trusted medical reference book.



4 Find Nearby Hospitals
Need care? Just tell your location and the assistant helps you find nearby healthcare facilities.



Key Features

Voice-based Interaction
Talk naturally on a phone call using Vapi.

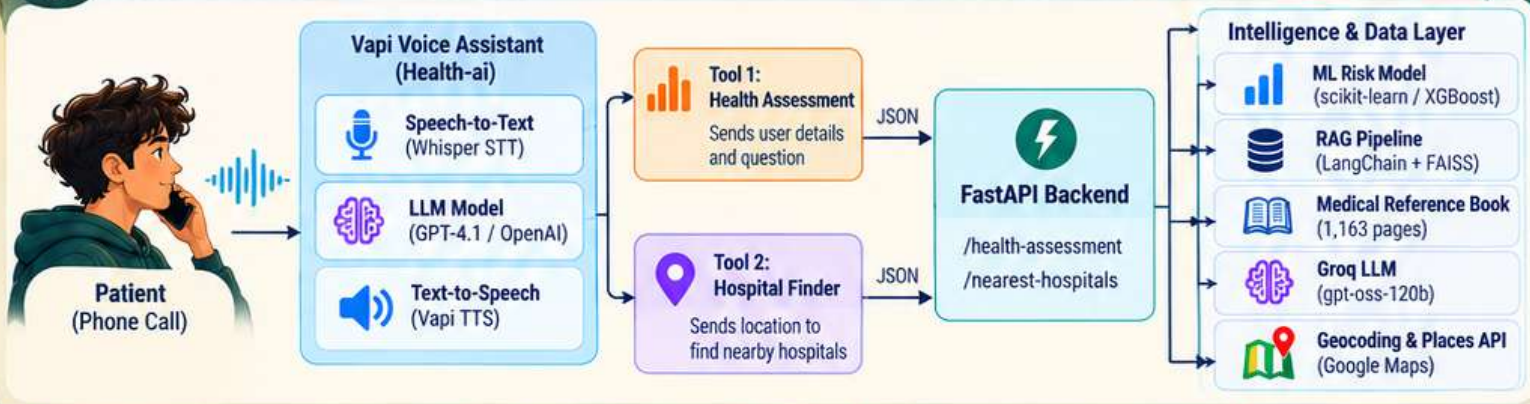
Health Risk Prediction
Predicts Low, Moderate or High risk using 9 lifestyle and medical inputs.

Reliable Health Information
Answers your health questions using a medical reference book (1,163 pages) with RAG.

Find Nearby Hospitals
Get nearby healthcare facilities from your spoken location.

System Architecture

Bringing AI, Voice and Medical Knowledge Together



Input Parameters (for Health Risk Assessment)

- Age
- Gender
- BMI
- Smoking Status
- Alcohol Consumption
- Physical Activity
- Sleep Duration
- Chronic Disease History
- Stress Level

Output

Health Risk Level
Low | Moderate | High

Personalized Explanation
Grounded in a medical reference book.

Nearby Hospitals
Name, address, distance and location details.

Technology Stack

- FastAPI (Backend)
- Python
- scikit-learn / XGBoost (Risk Model)
- LangChain (RAG Pipeline)
- FAISS (Vector Index)
- Groq (LLM for RAG)
- Vapi (Voice Assistant)
- Google Maps APIs (Gecooding & Places)

Impact

- Easy access to preliminary health guidance
- Grounded and reliable information
- Helps find nearby healthcare facilities
- Encourages proactive health awareness

Towards Healthier Communities

This system is a screening and information tool. It does not diagnose disease, prescribe treatment or replace a qualified clinician.

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